Programming with R Project

Paris Restaurant Recommendation System

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Introduction

The Restaurant Recommendation System is a Shiny-based web application that allows users to explore restaurants in Paris based on different criteria. It offers several interactive features, including filtering by cuisine category, subcategory, price level, rating range, and dietary preferences (e.g., vegetarian-friendly and gluten-free). Additionally, the system displays a map to visualize the restaurant locations and a table to review the filtered results. Users can also sort the restaurants by ratings, popularity, or review count.

Features & Functionality

1. **Sidebar Filters**:
    - **Cuisine Category**: Users can choose a cuisine category (e.g., European, Asian, etc.) to filter restaurants based on their culinary style.
    - **Cuisine Subcategory**: After selecting a category, the user can narrow down their choice further by selecting a subcategory (e.g., Italian, Japanese, etc.).
    - **Price Level**: The user can choose from different price ranges using a dropdown. The price levels are classified as:
        - € (Low)
        - €€-€€€ (Moderate)
        - €€€€ (High)
    - **Rating Range**: A slider is provided for users to filter restaurants based on average rating. They can set a minimum and maximum rating value.
    - **Dietary Filters**: Three checkboxes are available for dietary preferences:
        - **Vegetarian-Friendly Only**: Filters for vegetarian-friendly restaurants.
        - **Gluten-Free Options Only**: Filters for gluten-free restaurants.
        - **Show Only Restaurants with Awards**: Filters for restaurants that have received awards.

    These filters allow users to customize their search results and find the restaurants that match their preferences.

2. **Map View**:
    - The app displays the location of restaurants on a **leaflet map**. Restaurants are marked with pins, and users can interact with the map to view more details about each restaurant, such as name, rating, price level, cuisine type, and awards.
    - The map allows for easy visualization of restaurant distribution across Paris.
3. **Table View**:
    - The filtered restaurants are displayed in a **table** format below the map. The table provides essential information such as:
        - Restaurant Name

        - Cuisine Category
        - Cuisine Subcategory
        - Average Rating
        - Price Level
        - Awards
    - The table is initially set to display the top 20 restaurants based on the filters. There is a **"Show More"** button that allows users to view more results if they are available.
4. **Sorting**:
    - Users can sort the restaurants by different criteria:
        - **Rating**: Sort restaurants by their average ratings.
        - **Popularity**: Sort restaurants by their popularity rank.
        - **Review Count**: Sort restaurants by the total number of reviews they have received.
    - Sorting helps users quickly find the best restaurants based on their preferences.
5. **Indicators**:
    - Above the map, the app displays several indicators:
        - **Total Restaurants**: The number of restaurants that match the selected filters.
        - **Average Rating**: The average rating of all filtered restaurants.
        - **Restaurants with Awards**: The number of filtered restaurants that have received awards.
6. **Responsive Layout**:
    - The app's layout is designed to be responsive and user-friendly. It uses **Shiny Dashboard** for a clean and professional look with a sidebar, a main content area, and tabs for different views (Map View and Table View).
    - Custom CSS has been applied to ensure that the app looks polished and visually appealing, with specific colors and spacing for the sidebar, map, and table.

How It Works

1. **Initial Load**: When the app first loads, it displays a clean, interactive sidebar with all available filters. The map view is displayed with restaurant pins for Paris, but no data is filtered until the user makes their selection.
2. **Filtering**: As the user makes selections in the sidebar, the app reacts immediately, filtering the restaurant data based on the user's criteria. For example:
    - If the user selects "European" for the cuisine category, the app will filter all restaurants to show only those that match this category.
    - Similarly, selecting a price level will filter the dataset to show only restaurants in the chosen price range.

3. **Map and Table Update**: Once filters are applied, the map and table are updated in real-time:
    - The **map** updates to show the selected restaurants' locations with popup information.
    - The **table** shows the filtered restaurants with the selected information (e.g., name, rating, price level, etc.).
4. **Show More**: The table initially shows the top 20 restaurants. If there are more than 20 filtered results, the **"Show More"** button becomes visible, allowing the user to load additional restaurants.
5. **Sorting**: The user can click on a column in the table to sort the results by rating, popularity, or review count. This helps the user prioritize which restaurants to explore based on their preferences.
6. **Indicators**: The indicators above the map update dynamically based on the filtered data:
    - The **Total Restaurants** indicator reflects the number of restaurants that match the selected filters.
    - The **Average Rating** indicator provides an overall rating of the filtered restaurants.
    - The **Restaurants with Awards** indicator shows how many of the filtered restaurants have received accolades.

Conclusion

This Restaurant Recommendation System provides an easy-to-use interface for users to filter and explore restaurants based on their preferences. It offers a map for visualizing restaurant locations, a table for viewing restaurant details, and various filters for a highly customizable experience. The app's design and interactivity ensure that users can make informed decisions when choosing a restaurant in Paris.

By combining Shiny's interactivity with a polished UI and clear visualization, this app offers an effective tool for restaurant recommendations.